SERATEGNA

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Master Specification v3.0 — Strategy, Architecture & Build Plan

The work-to-credit operating system for Ethiopia's real economy. This document reorganizes the full Serategna concept around a single principle: the marketplace must win before the score can matter. It integrates every v2.0 safeguard (Fayda identity, escrow without stored value, SoS safety, guarantors, Score v2.0, back-testing) and applies four structural corrections: Phase 1 carries the most weight, identity verification becomes a tiered gate instead of a wall, the Nisir relationship is governed before it is used, and the payment rail is sequenced through partners that will actually sign with a startup.

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How This Document Is Organized

Part A sets the strategic foundation: the thesis, what changed from v2.0 and why, the Ethiopian business-culture playbook, and the full legal map. Parts B, C and D are the three phases — each with its own strategy, product specification, technical architecture, and legality/culture checkpoints, so each phase can be handed to a build team as a self-contained brief. Part E is the consolidated engineering plan: stack, data model, security, sprint roadmap, risk register, and the next 30 days.

Three recurring callout types appear throughout: **Legality** (amber) flags regulatory requirements to verify with Ethiopian counsel; **Culture & practice** (purple) encodes how business is actually done in Ethiopia; **Design note** (green) records a deliberate product decision.

PART A — Strategic Foundation

A1. The Thesis (Unchanged) and the Correction (New)

The thesis stands: Ethiopia's informal economy — roughly 85% of employment — generates billions of birr of real work with zero financial visibility. Serategna converts verified, paid work into a portable financial identity. The marketplace is the customer-acquisition cost; the proprietary earnings dataset and the Serategna Score are the asset; financial-services margin is the business. Each phase generates the asset the next phase monetizes.

The correction is one of emphasis and sequencing. Earlier versions spent most of their depth on the scoring engine — the part that can be built — and treated marketplace density — the part that must be won — as a given. This document inverts that. Phase 1 is the longest part of this specification because the entire business dies in months 0–12 if workers and clients meet once on Serategna and then transact in cash forever. Every Phase 1 feature is therefore justified against one question: does it keep the transaction on the platform?

Recommendation applied	What changes in this version	Where specified
Phase 1 carries the weight	Disintermediation defense package: escrow guarantees, day-one visible Score, worker loyalty economics, delala agent program, rebooking convenience	Part B, sections B1–B2
Fayda as tiered gate, not wall	Tier 0 onboarding on phone + Telebirr KYC with escrowed payouts; full payout release and credit eligibility require Fayda (Tier 1)	Part B, section B2.2

Nisir governance first	Formal conflict-of-interest protocol: disclosure, recusal, arm's-length pricing, parallel second-lender track	Part C, section C1
Payment rail realism	Launch on a licensed aggregator (Chapa-class PSO); direct Telebirr/CBE Birr commercial integrations pursued in parallel, not assumed	Part B, section B3.4

A2. How Business Is Actually Done in Ethiopia — The Operating Assumptions

Serategna does not import a Western gig-app playbook; it digitizes Ethiopian trust mechanics. Every assumption below shapes a concrete feature decision later in this document.

- **Delala (brokers) are the incumbent network.** Home services, housing, and labor in Addis flow through commission-based brokers operating on reputation and personal networks. Competing with delalas head-on is slow and culturally tone-deaf; converting them is fast. Serategna recruits high-reputation delalas as licensed 'Serategna Agents' who earn a digital referral commission on every worker they onboard and a declining share of those workers' first-six-month transactions. The broker keeps earning; the platform gets distribution, vetting knowledge, and neighborhood trust it cannot buy with ads.
- **Trust is personal, not documentary.** An Ethiopian household hires the cleaner their neighbor vouches for, not the highest-rated stranger. Product implications: referral chains are first-class data ('vouched by'), guarantor-backed lending mirrors how families already stand behind borrowers, and ratings display the rater's neighborhood and repeat-hire status rather than anonymous stars.
- **Negotiation is the norm; fixed pricing is foreign.** Most service prices in Ethiopia are haggled. The bidding flow is therefore not a feature but the culturally native pricing mechanism. Serategna publishes a fair-price guidance band per job category and sub-city (built from its own transaction data over time) and lets parties negotiate within the app — capturing the haggle instead of being bypassed by it.
- **Cash is king; digital payment needs a reason.** Telebirr adoption is broad but cash remains the default for informal services. Paying through the app must buy the client something cash cannot: escrow protection (money released only on confirmed completion), a rework guarantee on in-app jobs, dispute resolution, and the SoS-verified safety layer. For the worker, in-app payment is the only path to the Score — and the Score must be visible and growing from the very first job.
- **Equb and iddir are the indigenous financial institutions.** Rotating savings groups (equb) and mutual-aid societies (iddir) already perform savings and insurance functions for millions. Phase 3 does not replace them; it digitizes them — verified-member digital equb circles and iddir-distributed micro-insurance — riding two centuries of social capital.

- **Religious composition shapes the credit product line.** A large share of the workforce is Muslim and will not take interest-bearing loans. Phase 2 must launch with a parallel interest-free track: murabaha (cost-plus) equipment financing and qard hasan-style earned-wage access. Nisir's existing IFFS directorate makes this the natural pilot structure rather than an afterthought.
- **The Ethiopian calendar and religious seasons drive demand.** The app supports dual calendars (Gregorian + Ethiopian) everywhere a date appears. Demand spikes around Meskel, Timket, Fasika, Eid and the wedding season (cleaning, catering help, repairs); fasting seasons reshape food-related work. The matching engine and the Score's seasonality adjustment must treat these as expected cycles, not volatility.
- **Language is identity.** Trilingual from day one — Amharic, Afaan Oromoo, English — with language-of-preference per user, not per app. Voice-note job descriptions are supported because literacy varies; an Afaan Oromoo-first experience is a genuine differentiator no competitor offers and aligns with where expansion (Adama, Jimma, wider Oromia) will go.
- **Devices and data are constrained.** Android dominates, devices are often low-RAM, and data is expensive. Hard budget: app under 30 MB, offline-first job queue, SMS fallback for every critical event, and a Telegram bot companion — Telegram is Ethiopia's de facto distribution channel and a zero-install entry point for both job posting and worker discovery.
- **Gender safety is a market-entry condition, not a feature.** Domestic work is dominated by women entering strangers' homes. The SoS system, client identity verification, female-worker-controlled job filters (e.g., verified-female-client-only visibility), and an emergency response protocol are prerequisites for the largest Phase 1 vertical to function at all.

A3. The Legal Map — Build Within These Lines

Serategna's structure is deliberately regulation-shaped: no stored value, no balance-sheet lending, partner-held risk. The table below is the working legal frame; every item must be confirmed by retained Ethiopian counsel before the corresponding feature ships.

Instrument	What it governs for Serategna	Design response
NBE Payment Instrument Issuer Directive (ONPS/01/2020 as amended)	Issuing stored value / wallets requires a PII license	No wallet. Client funds sit in Serategna's corporate escrow sub-account at CBE; the in-app 'balance' is a ledger view of money already owed, withdrawable only to the worker's own Telebirr/bank account
NBE Payment System Operator licensing	Operating payment switching/gateway services requires a PSO license	Launch through an already-licensed PSO/aggregator (Chapa-class);

		Serategna is a merchant of record for its commission only, never a payment operator
NBE lending & MFI regulation	Lending is reserved for licensed banks/MFIs	All credit originates on the partner MFI/bank balance sheet; Serategna sells scoring data and origination services, never credit
Personal Data Protection Proclamation No. 1321/2024	Consent, purpose limitation, data-subject rights, breach notification, restrictions on cross-border transfer	Data residency in Ethiopia; explicit, per-purpose, revocable consent for sharing the Score with each named lender; full data-subject access/erasure tooling; DPO appointed before Phase 2
Fayda — National ID Proclamation No. 1284/2023	Legal basis and procedure for digital ID verification	Tiered verification (B2.2); MOSIP API integration when access is granted; manual document review as the interim path
Commercial Code (Proclamation No. 1243/2021)	Company formation; suretyship/guarantee enforceability	Serategna PLC incorporation; guarantor agreements drafted to the Code's suretyship requirements, with amount-capped digital guarantees
Electronic Transactions Proclamation No. 1205/2020	Validity of electronic contracts and signatures	OTP-based digital assent for ToS, job contracts and small guarantees; wet-ink or in-branch confirmation retained for guarantees above a counsel-set threshold until enforceability is court-tested
Labour Proclamation No. 1156/2019	Employee vs independent-contractor boundary	Workers are independent contractors: they set availability, accept or reject freely, work multi-platform, supply own tools; Serategna never sets schedules or exclusivity. Misclassification is a standing item on the legal risk register
Tax law (VAT/TOT, income tax)	VAT on platform commission; workers' own income-tax position	VAT charged on the take-rate only (not gross job value); annual earnings statements issued to workers — a feature, since it is the first income proof

		many have ever had; withholding obligations confirmed with counsel
Movable Property Security Right Proclamation No. 1147/2019	Movable collateral registration	Enables Phase 2/3 equipment financing secured on the financed tools themselves, registered in the collateral registry by the partner lender

> **Legality:** Engagement of a retained Addis Ababa law firm with NBE and data-protection practice is a Month-0 action, not a Phase 2 one. The two questions needing written opinions earliest: (1) confirmation that the CBE escrow sub-account structure does not constitute stored-value issuance; (2) enforceability conditions for OTP-signed guarantor agreements.

PART B — Phase 1: The Marketplace (Months 0–12)

B1. Phase 1 Strategy: Win Density, Defeat Leakage

Goal: become the default way two verticals transact in Addis Ababa — home services (cleaning, plumbing, electrical, repairs) and delivery/errands — measured not in signups but in verified paid jobs per worker per week. The single existential risk is disintermediation: a client and worker meet once through the app, exchange phone numbers, and move to cash. Every incentive in the informal economy points that way, and the credit reward that ultimately solves it does not activate until a worker has months of history. Phase 1 therefore deploys a layered defense that makes staying on-platform rational for both sides from job one:

1. **Client-side value that cash cannot offer.** Escrow (money released only on confirmed completion), a rework guarantee on in-app-paid jobs (free return visit within 72 hours, funded by a 1% guarantee reserve from the take-rate), dispute mediation with a 48-hour SLA, verified worker identity, and the SoS-audited safety record. Off-platform, the client loses all five the moment they hand over cash.
2. **Worker-side value visible from day one.** The Score begins accruing on the first paid job and is displayed prominently with a projection ('at this pace you reach Credit-Eligible in March'). Plus immediate loyalty economics: airtime/data rewards on weekly job streaks, free annual earnings statements (first formal income proof of their lives), priority dispatch ranking tied to on-platform volume, and partner discounts on tools and supplies that scale with verified history.
3. **Rebooking friction lower than a phone call.** One-tap rehire of a past worker, saved addresses, recurring job scheduling (every-Saturday cleaning), and family accounts. The diaspora wedge belongs here: a relative in Washington or Dubai can book and pay for services for family in Addis — a use case where cash is impossible and Serategna has no competitor.
4. **The delala agent program.** Brokers are paid to fill the platform rather than to route around it: onboarding commission per verified worker, plus a declining transaction share for six months. Agents also serve as the physical verification touchpoint (B2.2) — Ethiopian trust networks become the KYC field force.
5. **Leakage detection and graduated response.** The system flags repeat client-worker pairs whose transactions stop after the first match. Response is incentive-first (a rebooking discount nudge to the pair), never punitive against workers — punishing leakage drives workers off entirely. Measured leakage is a board-level KPI; the target is keeping repeat-pair on-platform payment above 60% by month 12.

> **Design note:** Take-rate launches at 7% on home services and 10% on delivery, fully visible to both sides, with the first three jobs per new client commission-free. Of the take-rate, one percentage point funds the guarantee reserve. Pricing below the perceived value of escrow + guarantee + safety is what makes the digital-payment ask culturally survivable.

B2. Phase 1 Product Specification

B2.1 Two-Sided Core

Capability	Worker side (Serategna)	Client side (households & businesses)
Onboarding	Phone + OTP; trade/category selection; voice or text bio; agent-assisted or self-serve; language choice (AM/OM/EN)	Phone + OTP; saved addresses; household or business profile
Discovery & matching	Geo-matched job feed (PostGIS radius + category + rating rank); instant-dispatch toggle for on-demand work	Post a job (text or voice note, photos); browse verified workers; see fair-price band for the category and sub-city
Pricing	Accept listed price or bid; counter-offer chat	Fixed-price or open-to-bids posting; in-app negotiation thread
Job lifecycle	Accept → travel (live location on) → start (geo-stamped) → complete (photo proof) → client confirms → escrow releases	Book → fund escrow → track arrival → confirm completion or open dispute within 24h
Ratings & trust	Two-way ratings; 'repeat client' and 'vouched-by' badges; public completion rate	Worker cards show Fayda-verified badge, neighborhood, repeat-hire count, guarantee eligibility
Money	Earnings ledger; withdraw to own Telebirr/bank; weekly statement; Score tracker with credit-eligibility projection	Pay via Telebirr/CBE Birr/card through licensed aggregator into escrow; receipts; refund on upheld dispute
Safety	SoS panic button on every active-job screen (GPS trail + encrypted audio + alert chain); silent-hold covert trigger; female-client-only visibility filter	Worker identity verified; live tracking of arrival; SoS coverage applies to clients on worker-premises jobs too

B2.2 Tiered Identity: Fayda Without the Funnel Cliff

Mandatory Fayda before any payment — the v2.0 rule — protects Phase 2 data quality but strangles Phase 1 density: coverage among informal workers is uneven and a 24-hour document review is a drop-off cliff for a labourer who needs work today. The tiered gate keeps both guarantees:

Tier	Requirement	Can do	Cannot do
Tier 0 — Provisional	Phone OTP + Telebirr account (which carries telecom KYC)	Full profile; accept up to 5 jobs or ETB 4,000 gross; earnings accrue in escrow ledger	Withdraw funds; appear as 'Verified'; accrue Score

Tier 1 — Fayda Verified	Fayda number + document/selfie review (24h, agent-assisted available) → MOSIP biometric API when access granted	Withdraw all escrowed earnings incl. Tier-0 backlog; Verified badge; Score accrual begins (Tier-0 jobs retro-credited)	Access credit products (Phase 2 gate)
Tier 2 — Credit-Ready	Tier 1 + ≥6 months history + consented data sharing + guarantor on file	Phase 2 products: earned-wage access, nano-loans	—

The escrowed-payout mechanic converts verification from a barrier into an incentive: by job three a Tier-0 worker has real money visible in the ledger that only Fayda verification unlocks. Expected effect: near-100% verification by the time meaningful earnings exist, with zero funnel loss at signup. Serategna Agents perform assisted verification in the field for workers without smartphone document-upload skills.

> **Legality:** Tier-0 reliance on Telebirr's underlying telecom KYC as provisional identity, and the retroactive Score-crediting of Tier-0 jobs, should both be confirmed against Proclamation 1284/2023 and NBE KYC expectations for the eventual lending partner.

B2.3 Telegram Companion

A Telegram bot mirrors the client posting flow (post a job, receive bids, pay via aggregator link) and gives workers job alerts and bid submission. It is the zero-download funnel: Ethiopia's gig labor market already lives in Telegram channels, and the bot meets it there, then graduates users to the app where escrow, SoS and the Score live. The bot shares the same API and ledger — a Telegram job is a first-class platform job.

B3. Phase 1 Technical Architecture

B3.1 Stack (continuous with the existing v1 codebase)

Layer	Technology	Notes
Mobile app	React Native (Expo), single app with worker/client mode switch	i18n AM/OM/EN; offline-first job queue (SQLite + sync); <30 MB budget; dual-calendar component
Web	React admin console + lightweight client web booking page	Admin: verification queue, disputes, SoS desk, agent management, finance ops
API	Node.js / Express, modular monolith with service boundaries (auth, profiles, jobs, ledger, payments, ratings, disputes, safety, notifications)	Boundaries drawn so ledger and (later) scoring extract cleanly into services

Data	PostgreSQL 16 + PostGIS; Redis (queues, matching, rate-limit)	Single primary + read replica; all money movements in an append-only double-entry journal
Payments	Aggregator adapter pattern: launch on a licensed PSO aggregator (Chapa-class) covering Telebirr, CBE Birr, cards; direct Telebirr/CBE commercial APIs added behind the same adapter when agreements land	Adapter interface: initiate, webhook-confirm, refund, payout
Messaging	FCM push + local SMS gateway fallback for every critical event; Telegram Bot API	SMS is the reliability floor on low-end devices
Maps/geo	Google Maps (existing integration) + PostGIS for matching radius and SoS trails	Cache tiles aggressively; data-cost aware
Infra	Docker (existing) on an Ethiopia-resident host for primary data, with encrypted offsite backup; CI via GitHub Actions; staging + prod	Data residency per Proclamation 1321/2024; latency to local users

B3.2 The Escrow Ledger — the Heart of Phase 1

Every birr is represented as immutable double-entry journal lines over internal accounts: client_funds_in_escrow, worker_payable, platform_commission, guarantee_reserve, refunds. A job's money lifecycle is a fixed state machine: FUNDED (aggregator webhook confirms client payment into the CBE escrow sub-account) → HELD (job in progress) → RELEASED (client confirmation or 24h auto-confirm) → split posted (worker payable + commission + reserve) → PAID_OUT (worker withdrawal to own account) — with DISPUTED and REFUNDED branches. Daily automated reconciliation matches the journal against the bank statement and aggregator settlement reports; any variance blocks payouts until resolved. This journal is simultaneously the compliance artifact (proving no stored value: every credit is owed money, withdrawable on demand) and the future Score's raw material — its integrity is a Phase 2 asset being built in Phase 1.

B3.3 Core Data Model (Phase 1 tables)

Table	Key fields / purpose
users	id, phone, role flags (worker/client/agent), language, tier (0/1/2), fayda_status, created via OTP auth
worker_profiles	categories[], bio (text/voice ref), service_radius, sub_city, availability, instant_dispatch, female_client_only_flag, vouched_by (agent/worker id)
jobs	client_id, category, description/media, location (PostGIS point), pricing_mode (fixed/bid), price_band shown, status state machine, geo-stamps (start/complete), photo_proof refs

bids / job_threads	worker_id, amount, message; negotiation history preserved (pricing intelligence corpus)
ledger_entries	append-only: txn_id, debit_account, credit_account, amount, currency, job_id, external_ref (aggregator/bank), posted_at; no UPDATE/DELETE grants
payouts	worker_id, destination (telebirr/bank), amount, status, external_ref
ratings	job_id, rater, ratee, stars, tags, text; repeat_pair computed
disputes	job_id, opener, reason, evidence refs, mediator_id, resolution, SLA timestamps
sos_events	job_id, trigger_type (button/silent), gps_trail (linestring), audio_ref (encrypted), alert_chain log, account_freeze actions
agents	delala identity, territory, onboarding_count, commission ledger link
verification_queue	user_id, doc refs, reviewer, decision, MOSIP txn id (later)

B3.4 Payment Rail Sequencing

Month 0–1: contract a licensed aggregator (evaluate Chapa, ArifPay, SantimPay, Kacha on payout API quality, settlement speed, and escrow-account compatibility — verify current licensing status with NBE's published PSO list). Months 3–9: pursue direct Telebirr enterprise integration via Ethio Telecom and a CBE Birr agreement in parallel — these reduce cost per transaction and settlement lag but must never be launch blockers. The adapter pattern means swapping or adding rails is a configuration change, not a rewrite.

B4. Phase 1 Legality & Culture Checkpoints

> **Legality:** Before first public transaction: PLC incorporated; trade license covering digital marketplace services; VAT registration; aggregator contract executed; CBE escrow sub-account opened with written bank acknowledgment of its client-funds purpose; ToS and worker contractor agreements reviewed against Labour Proclamation 1156/2019; privacy policy and consent flows compliant with 1321/2024; SoS protocol agreed with a private emergency-response provider and aligned with police-contact procedure.

> **Culture & practice:** Pilot launch is relational, not digital: two sub-cities (recommend Bole and Yeka for client density), launched through 20–30 recruited delala agents, neighborhood iddir presentations, and church/mosque-adjacent community boards — before any paid digital marketing. The first 500 workers should be hand-verified and personally known to agents; their service quality is the brand.

B5. Phase 1 KPIs and Exit Criteria

KPI (monthly board metric)	Target by Month 12

Verified, paid jobs per active worker per week	≥ 2.5
Active Tier-1 verified workers	≥ 3,000
Cumulative completed paid jobs	≥ 100,000
Repeat-pair on-platform payment share (leakage inverse)	≥ 60%
Client repeat-booking rate (90-day)	≥ 40%
Dispute rate / resolution within SLA	< 3% of jobs / > 95% in 48h
Ledger reconciliation breaks	Zero unresolved > 24h
Workers with ≥6 months continuous history (Phase 2 fuel)	≥ 1,200

Phase 2 does not begin on a calendar date; it begins when the last row of this table is true, the MFI partnership is signed, and the data-protection registration and consent infrastructure are live.

PART C — Phase 2: Financial Services (Months 13–24)

C1. Nisir Relationship Governance

Nisir MFI is the natural first lending partner: shared institutional familiarity, existing IFFS (Islamic Financial and Financing Services) directorate for the interest-free track, and an understanding of the informal-sector borrower. The relationship also carries a conflict-of-interest risk that must be governed before a single birr of credit is originated, not after.

The governance protocol has four standing elements:

1. **Disclosure at board level.** Serategna's founding documents and its MFI partnership agreement both disclose the relationship explicitly. Any Nisir board or management personnel with an ownership or advisory stake in Serategna recuse from decisions touching the Serategna contract, scoring methodology, pricing, or data-sharing terms.
2. **Arm's-length commercial terms.** The data-licensing and origination-fee agreement is benchmarked against terms Serategna could obtain from a second, unaffiliated MFI. A term sheet from at least one independent lender (Addis Credit & Savings Institution or similar) is obtained before the Nisir agreement is signed — this establishes the market rate and demonstrates the arrangement's independence.
3. **Independent credit decision.** Serategna delivers the Score and the consented data package; Nisir makes all credit decisions independently, applying its own credit policy. Serategna has no visibility into individual Nisir credit decisions and no economic interest in approval rates — its fee is per consented data share, not per approved loan.
4. **Periodic review.** The partnership agreement is reviewed annually by both boards, with pricing benchmarked to market. Either party may terminate on 90 days' notice. The review is a standing agenda item for Nisir's Audit & Risk Committee.

> **Legality:** The data-sharing arrangement requires a Data Processing Agreement under Proclamation 1321/2024 specifying Nisir as a data processor (not a controller) for Serategna worker data. Individual workers must consent to each named lender before any data is shared — a blanket 'we may share your Score with MFI partners' clause is not adequate under purpose-limitation requirements.

> **Design note:** The second-lender track (pursuing at least one non-Nisir MFI partnership simultaneously) is not a backup plan; it is the governance mechanism. A functional alternative lender at arm's length proves the Nisir terms are commercial and gives the Serategna Score independent market validation.

C2. Phase 2 Credit Products

Product	Structure	Size / term	Eligibility gate
Earned-Wage Access (conventional)	Advance on confirmed, escrowed, unreleased earnings. Not a loan — no	Up to 80% of escrowed balance;	Tier 1; ≥2 completed paid jobs this week; no open dispute

	interest. Platform fee: flat ETB 15 per advance.	max ETB 2,000 per advance	
Earned-Wage Access (qard hasan)	Same structure, no fee. Funded from a sadaqah pool seeded by platform and willing clients. Waitlist managed by queue.	ETB 500 max; 7-day repayment	Tier 1; Muslim self-identification; 0 prior defaults
Nano-loan (conventional)	Cash advance against 90-day earnings trajectory. Originated on Nisir balance sheet. Serategna fee: origination referral.	ETB 1,000–5,000; 30–90 days; rate per Nisir credit policy	Tier 2; ≥6 months history; Score ≥ 620; guarantor signed
Nano-loan (murabaha)	Nisir purchases a tool/input and sells to worker at cost-plus markup, paid back from earnings. Asset title held by Nisir until paid.	ETB 2,000–10,000; 3–6 months	Tier 2; ≥6 months; Score ≥ 580; halal asset category confirmed
Equipment micro-lease	Third-party lessor (not Serategna, not Nisir). Serategna refers and guarantees via Score attestation.	Asset-value dependent; 6–24 months	Tier 2; Score ≥ 650; 9 months history

> **Design note:** Earned-wage access on escrowed (not yet released) earnings is not a loan product under NBE MFI regulations — it is Serategna advancing its own ledger obligations early. This interpretation must be confirmed in writing by retained counsel and by NBE's Legal Affairs directorate before launch.

C3. The Serategna Score v2.0

C3.1 Score Architecture

The Score is a single number 300–850, updated weekly, displayed to the worker at all times and shared with consenting lenders on request. It has four components:

Component	Weight	What it measures
Transaction integrity	35%	On-platform payment share (vs inferred cash), dispute rate, dispute outcomes, chargeback/refund history. The core disintermediation signal.
Earnings consistency	30%	Weekly earnings stability and growth trend; seasonality-adjusted (Ethiopian calendar cycles treated as expected, not as volatility); multi-category diversification bonus.

Platform behavior	20%	Completion rate, acceptance rate, response time, client ratings, repeat-hire rate, streak maintenance, profile completeness.
Relationship capital	15%	Guarantor quality (their Score and MFI history), iddir/equb membership verification (Phase 3), vouched-by chain depth.

C3.2 Score Integrity and Back-Testing

Back-testing protocol: before any Score-linked lending begins, Serategna runs a retrospective simulation over 6–12 months of Phase 1 ledger data, computing what scores workers would have had and comparing predicted creditworthiness against observed behavioral proxies (dispute rate, leakage behavior, platform retention). A Score that cannot distinguish good from bad platform behavior has no predictive power for credit and must be recalibrated before Phase 2 launches.

Live monitoring: once lending begins, Score v2 is calibrated quarterly against 30-day and 90-day loan performance from Nisir and the second lender. If a Score cohort's default rate exceeds the predicted range by more than one standard deviation, the component weights are reviewed by the joint Serategna-Nisir data committee, with the recalibrated model applied prospectively only.

> **Legality:** Sharing the Score with lenders constitutes sharing personal financial data. Each share event requires (a) the worker's explicit, revocable, named-lender consent, (b) a log entry timestamped to the second, (c) a data subject access mechanism allowing workers to see exactly what was shared and when, and (d) a right-of-erasure process that removes the shared record without affecting the worker's own Score history.

C4. Phase 2 Technical Additions

Addition	Specification
Score engine	Python microservice (extracted from the monolith at Phase 2 boundary): reads ledger and behavioral tables, applies component weights, writes score_history (worker_id, score, components JSON, computed_at, model_version). Exposes internal API: GET /score/{worker_id}?as_of={date}
Consent ledger	consent_events table: worker_id, lender_id, scope (score_only / full_data_package), granted_at, revoked_at. No data share fires without a live, unrevoked consent row.
Lender API	Authenticated REST endpoint: POST /lender/score-request (lender credentials + worker consent token → score + consented data package JSON). Rate-limited; audit-logged.
DPO tooling	Admin module: data-subject access report generation (all data held for a given phone/Fayda ID); erasure workflow with confirmation and audit trail; breach-notification template with 72h clock.

EWA disbursement	Serategna advances from its own escrow sub-account to the worker's Telebirr; the ledger records the advance and the auto-repayment when the job escrow releases. No third-party lender involvement for EWA.
Guarantor module	Digital guarantor agreements (OTP-signed, counsel-reviewed threshold); guarantor dashboard showing their exposure, repayment status; automated alert at 7 days before maturity.

C5. Phase 2 KPIs and Exit Criteria

KPI	Target by Month 24
Active Tier-2 credit-eligible workers	≥ 800
EWA advances disbursed (cumulative)	≥ 5,000 advances
Nano-loan portfolio (Nisir + second lender)	ETB 2M+ outstanding
30-day nano-loan default rate	< 8%
Score predictive validity (Gini coefficient)	> 0.35 on back-test dataset
Data-subject consent rate (workers sharing Score)	> 70% of Tier-2 workers
Zero data breaches / regulatory findings	0

PART D — Phase 3: Platform Economy (Months 25–36)

D1. Phase 3 Strategy: Monetize the Asset

By Month 24, Serategna holds a proprietary dataset that no bank, MFI, or fintech in Ethiopia possesses: a multi-year, verified, granular earnings record for thousands of informal workers, paired with behavioral and social-capital data. Phase 3 monetizes this asset across three vectors while deepening the worker's financial life.

- **Lender diversification.** Open the Score API to two additional licensed MFIs or microfinance banks on identical arm's-length terms. The Score becomes a multi-lender utility; competition among lenders improves rates for workers and increases Serategna's origination-fee revenue.
- **Digital equb.** Verified Serategna workers form digital rotating savings circles. The platform handles contribution scheduling, payout rotation (randomized or consensus-voted), and dispute resolution. Equb circles of platform-verified members have zero adverse-selection risk compared to informal cash circles — the Score history is the membership credential. Monetization: a flat monthly circle-management fee (ETB 20/member/month) and an optional interest-bearing (or murabaha) bridge advance to cover a member who misses a contribution cycle.
- **Insurance distribution.** Partner with a licensed Ethiopian insurer to offer iddir-style micro-insurance (accident, illness, tool loss) priced against the worker's earnings history. Serategna earns a distribution commission; the insurer gains a verified low-adverse-selection customer base it cannot acquire otherwise. Phase 3 does not hold insurance risk.
- **Business accounts.** Frequent clients (households and SMEs posting 5+ jobs/month) are offered a Business Account: bulk booking, invoicing, team access, and a verified spend report (useful for expense accounting). Take-rate on business accounts drops to 5% in exchange for volume commitment and prepaid credit.
- **Diaspora remittance-to-service.** Formalize the diaspora use case (Phase 1 wedge) as a product: diaspora accounts in a foreign currency, with automatic conversion and service booking on behalf of family in Ethiopia. Partnership with a licensed remittance operator (compliant with NBE's cross-border rules) handles the FX leg.

D2. Phase 3 Product and Technical Additions

Feature	Specification
Multi-lender Score API	Existing lender API extended with lender-tier pricing (volume-based discount schedule); lender onboarding workflow (AML/KYC of lender institution, API credential issuance, data-use agreement signing).
Digital equb module	equb_circles, equb_members, equb_contributions, equb_payouts tables; contribution reminder push/SMS 48h and 2h before due; payout

	state machine; dispute escalation to platform mediator; circle Score floor (minimum member Score to join a circle).
Insurance API bridge	Webhook integration with partner insurer's policy API: premium quote (based on earnings history with consent), policy issuance, claim notification. Serategna stores policy_id and premium_history; does not store health or claims data beyond notification.
Business accounts	Organisation entity above user; team member roles (booker, finance, admin); consolidated invoice generation (PDF, VAT-compliant); prepaid credit ledger; volume-tier take-rate engine.
Diaspora product	Foreign-currency account (passthrough, held at partner remittance operator); booking-on-behalf flow with beneficiary address book; FX rate display; confirmation to diaspora sender when job completes.
Score v3 additions	Equb membership and contribution history added to Relationship Capital component (weight stays 15%, sub-components rebalanced); insurance claim history (with consent) added as a negative-signal input to Transaction Integrity.

D3. Phase 3 Legal Additions

> **Legality:** Insurance distribution requires an insurance intermediary (agent or broker) license from the Ethiopian Insurance Authority, separate from NBE. Diaspora remittance handling requires either a sub-agent agreement under a licensed remittance operator or a dedicated cross-border payment service license — confirm the sub-agent route's scope with NBE's Foreign Exchange Supervision Directorate before product design is finalized.
>
> **Legality:** Digital equb with a bridge-advance feature may constitute a deposit-taking or credit activity depending on how the advance is structured. If Serategna funds the bridge from its own balance sheet, it must be structured as a merchant advance (against future platform earnings) to stay within the no-lending-license design constraint. Counsel opinion required before this feature ships.

D4. Phase 3 KPIs

KPI	Target by Month 36
Lender partners (active Score API users)	≥ 3
Active digital equb circles	≥ 200
Workers covered by micro-insurance	≥ 2,000
Business accounts	≥ 150
Annualized gross revenue run rate	ETB 18M+
Annualized origination-fee revenue (from lenders)	ETB 4M+

Platform net margin (before owner distributions)	> 20%

PART E — Engineering Plan

E1. Security Architecture

Security is not a Phase 2 feature; the escrow ledger holds real money from Month 1. The following controls are baseline requirements before the first public transaction:

- **Authentication.** Phone OTP via a reliable local SMS gateway (not international aggregator — Ethiopian networks have delivery issues with foreign origination). JWT access tokens (15-minute expiry) + refresh tokens (30-day, single-use rotation). Admin console requires TOTP 2FA.
- **Transport.** TLS 1.3 everywhere. Certificate pinning in the mobile app. HSTS on all web surfaces.
- **Ledger immutability.** The ledger_entries table has no UPDATE or DELETE grants at the database role level. All corrections are new journal entries with a reference to the corrected entry. Append-only enforced at both application and database layers.
- **Encryption at rest.** PII fields (phone, Fayda number, document refs, audio SoS recordings) encrypted at the application layer with a key held in a separate secrets manager (not in the database). Encryption keys rotated annually.
- **SoS audio.** Encrypted immediately on upload with a public key whose private key is held only in the emergency-response provider's HSM. Serategna cannot decrypt SoS audio — only the emergency coordinator can. This is a trust feature, not just a security one.
- **Payments.** Aggregator webhooks validated by HMAC signature. Payout requests require a second-factor confirmation from the worker (OTP to registered phone). Daily payout cap per worker (configurable, default ETB 10,000) with manual review above cap.
- **Access control.** Role-based access in the admin console: agent, operations, finance, trust-and-safety, engineering, super-admin. All admin actions are audit-logged (who, what, when, on which record). Quarterly access review.
- **Vulnerability management.** Dependency scanning in CI (npm audit, Dependabot). No public APIs without authentication. Rate limiting on all endpoints (Redis token bucket). Penetration test before Phase 2 launch by an independent firm.

E2. Sprint Roadmap

Sprint	Weeks	Deliverables
S1	1–2	Legal: law firm retained, PLC incorporation initiated, aggregator evaluation shortlist. Infra: staging environment on Ethiopia-resident host, CI pipeline, secrets manager. Auth: phone OTP flow, JWT + refresh, user table.
S2	3–4	Worker profile creation (categories, bio, language, availability). Client job posting (text, voice note, photos, location pin). PostGIS matching: radius + category query.

S3	5–6	Bid/negotiation thread. Job lifecycle state machine (accept → travel → start → complete → confirm). Geo-stamps. Photo proof upload.
S4	7–8	Aggregator integration (first PSO): payment initiation, webhook confirm. Escrow ledger double-entry journal. Escrow FUNDED → HELD → RELEASED state machine. Automated reconciliation scaffold.
S5	9–10	Ratings (two-way, tags). Dispute flow (open, evidence, mediator queue, resolution). Two-way notification (FCM + SMS fallback). Payout to Telebirr/bank.
S6	11–12	SoS panic button + silent trigger. GPS trail recording. Emergency alert chain. Account-freeze on trigger. SoS desk in admin console.
S7	13–14	Tiered identity: Tier-0 cap enforcement, verification queue, Tier-1 upgrade, escrowed-backlog release. Agent onboarding module and commission ledger.
S8	15–16	Score v1: compute weekly from ledger + behavioral data. Score display on worker dashboard with credit-eligibility projection. Score history table.
S9	17–18	Telegram bot: job posting, bid notification, payment link. Bot ↔ API shared ledger. Dual-calendar component. Fair-price band display (seeded with pilot data).
S10	19–20	Admin console: verification queue, dispute management, SoS desk, agent management, ledger ops, reconciliation dashboard. Load and security testing.
S11	21–22	Soft launch: 20–30 agents, 500 workers, Bole + Yeka. Monitored transaction flow. Reconciliation live. Bug fixes and UX iteration from real usage.
S12	23–24	Phase 1 stabilization: leakage detection and nudge system, rebooking UX, family accounts, diaspora booking flow. Phase 1 KPI dashboard.
Phase 2 prep	25–28	Back-test Score v2 on Phase 1 ledger data. Negotiate Nisir data agreement + second-lender term sheet. Consent infrastructure. DPO tooling. Lender API design.
Phase 2 launch	29–36	EWA (conventional + qard hasan). Score v2 live. Nano-loan referral flow. Guarantor module. Phase 2 KPI tracking.

E3. Risk Register

Risk	Likelihood	Impact	Mitigation

Disintermediation — workers/clients go to cash after first match	High	Existential	Full B1 defense package. Leakage KPI tracked weekly. Take-rate only charged after escrow confirms; cash transactions generate no take-rate but also no platform cost.
Fayda API access delayed or restricted	Medium	High (Phase 2 gate)	Tiered identity design means Phase 1 runs on Telebirr KYC + manual review. Phase 2 back-tested before Fayda dependency is critical. Manual path remains available indefinitely.
Aggregator PSO fails or deregisters	Low–Medium	High (payments stop)	Adapter pattern: swap aggregator in < 1 sprint. Dual-aggregator contract from Month 6 if volume justifies. CBE escrow sub-account holds funds independently of aggregator.
Nisir conflict-of-interest regulatory finding	Low	High (license risk for Nisir)	Governance protocol (C1) implemented before any credit originates. DPA and consent infrastructure precede all data sharing. Legal opinion obtained.
SoS incident with inadequate response	Low	Reputational/legal	Emergency-response provider contract in place before launch. Protocol tested quarterly. Female-client-only filter reduces exposure in highest-risk jobs. Incident review mandatory within 24h of any SoS trigger.
Data breach of PII or SoS audio	Low	High (regulatory + trust)	Application-layer encryption, append-only ledger, role-based admin access, annual pen test. Breach-notification playbook (72h to regulator per 1321/2024). SoS

			audio isolated and not accessible to Serategna staff.
Labour misclassification ruling	Medium	Medium (cost, redesign)	Worker agreement drafted against Labour Proclamation 1156/2019 with counsel. No exclusivity, no schedule-setting, own-tools provision. Monitored on legal risk register; worker status reviewed if law changes.
Score gaming by workers inflating on-platform signals	Medium	Medium (credit quality)	Score incorporates dispute outcomes, leakage detection, and client ratings — gaming requires coordinating multiple parties. Lender default data provides the ground truth that exposes gaming at the cohort level. Quarterly recalibration.
Low smartphone penetration in target demographics	Medium	Medium (reach cap)	Telegram bot as zero-download channel. SMS fallback for all critical events. Agent-assisted onboarding. Offline job queue. Dual-SIM / low-RAM Android optimization.
Ethiopian macro shock (FX, inflation, political)	Medium	Medium–High	All fees in ETB; no FX exposure until Phase 3 diaspora product. Nano-loan terms short (30–90 days) — portfolio re-prices quickly. No long-dated balance-sheet risk.

E4. The Next 30 Days — Month-0 Action List

These actions are not sprint tickets; they are institutional prerequisites. Nothing ships publicly until every item below is complete.

1. **Retain Ethiopian counsel** with NBE, data-protection, and commercial practice. Mandate: (1) written opinion on CBE escrow sub-account vs stored-value issuance, (2) OTP-signed guarantor enforceability opinion, (3) labour contractor agreement review, (4) privacy policy and consent-flow compliance review against Proclamation 1321/2024.
2. **Initiate Serategna PLC incorporation** at the Ethiopian Investment Commission / Ministry of Trade. Concurrent trade license application for digital marketplace services. VAT registration initiated.
3. **Open CBE escrow sub-account.** Obtain written bank acknowledgment that the account holds client funds in trust and is not a Serategna operating account. This letter is the compliance artifact for the no-stored-value position.
4. **Evaluate and shortlist payment aggregators.** Request API documentation, settlement timelines, payout capabilities, and NBE PSO license confirmation from Chapa, ArifPay, SantimPay, and Kacha. Target: signed aggregator agreement by end of Month 1.
5. **Recruit 5–10 pilot delala agents** in Bole and Yeka sub-cities. These are individuals known to the founding team or to community leaders, with existing worker networks. They are not employees; they sign the agent agreement (reviewed by counsel) and receive a physical orientation on the product, the verification process, and the SoS protocol.
6. **Set up engineering infrastructure:** Ethiopia-resident cloud host (evaluate Ethio Telecom Cloud, Safaricom Cloud Ethiopia, or a local data center with international backbone), staging environment, GitHub repository and CI pipeline, secrets manager, and monitoring stack (Prometheus + Grafana or equivalent).
7. **Begin NBE engagement.** Request an informal dialogue meeting with NBE's Financial Inclusion and FinTech directorate to present the Serategna model and the escrow/no-wallet structure. Not a license application — a relationship-building meeting to surface regulatory questions early. Ethiopian regulatory practice rewards proactive disclosure.
8. **Begin Nisir governance negotiation.** Initiate the data-partnership term sheet discussion with Nisir management. Simultaneously request a term sheet from a second MFI. Target: both term sheets in hand before any Phase 2 feature is built.

E5. Revenue and Unit Economics Summary

Revenue stream	When live	Economics
Marketplace take-rate (7–10% of job value)	Phase 1, Month 3	Primary revenue. At ETB 300 avg job value, 7% = ETB 21/job. At 100K jobs/year = ETB 2.1M gross.
Guarantee reserve (1% of job value)	Phase 1, Month 3	Reserve fund, not P&L revenue. Net to Serategna only to the extent reserves exceed claims.

Score licensing / origination referral fee	Phase 2, Month 15	ETB 150–300 per loan originated via Serategna Score. At 5,000 loans/year = ETB 750K–1.5M.
EWA flat fee (ETB 15/advance)	Phase 2, Month 15	At 10,000 advances/year = ETB 150K.
Equb circle management (ETB 20/member/month)	Phase 3, Month 28	At 2,000 circle members = ETB 480K/year.
Insurance distribution commission (~15% of premium)	Phase 3, Month 30	At ETB 500 avg annual premium, 2,000 workers = ETB 150K commission.
Business account take-rate (5%, higher volume)	Phase 3, Month 28	Incremental; structured as volume-commit offset against margin compression.
Diaspora service fee (flat ETB 50/booking)	Phase 3, Month 32	At 5,000 diaspora bookings/year = ETB 250K.

Blended 36-month revenue projection (Phase 1–3 combined, conservative case): ETB 15–20M cumulative gross revenue. The business becomes cash-flow-positive in Month 18–20 if Phase 1 KPIs are met on schedule, driven by take-rate volume before Score licensing activates. The Score licensing and equb lines are the high-margin, low-marginal-cost revenue streams that drive the long-run economics; marketplace take-rate funds the path to get there.

> **Design note:** This document ends here. The next artifact in the build sequence is the technical data model ERD (Phase 1 tables in detail), followed by the agent onboarding kit (field guide + agreement template) and the worker welcome pack (Amharic, Afaan Oromoo, English — voice-over versions for low-literacy users).